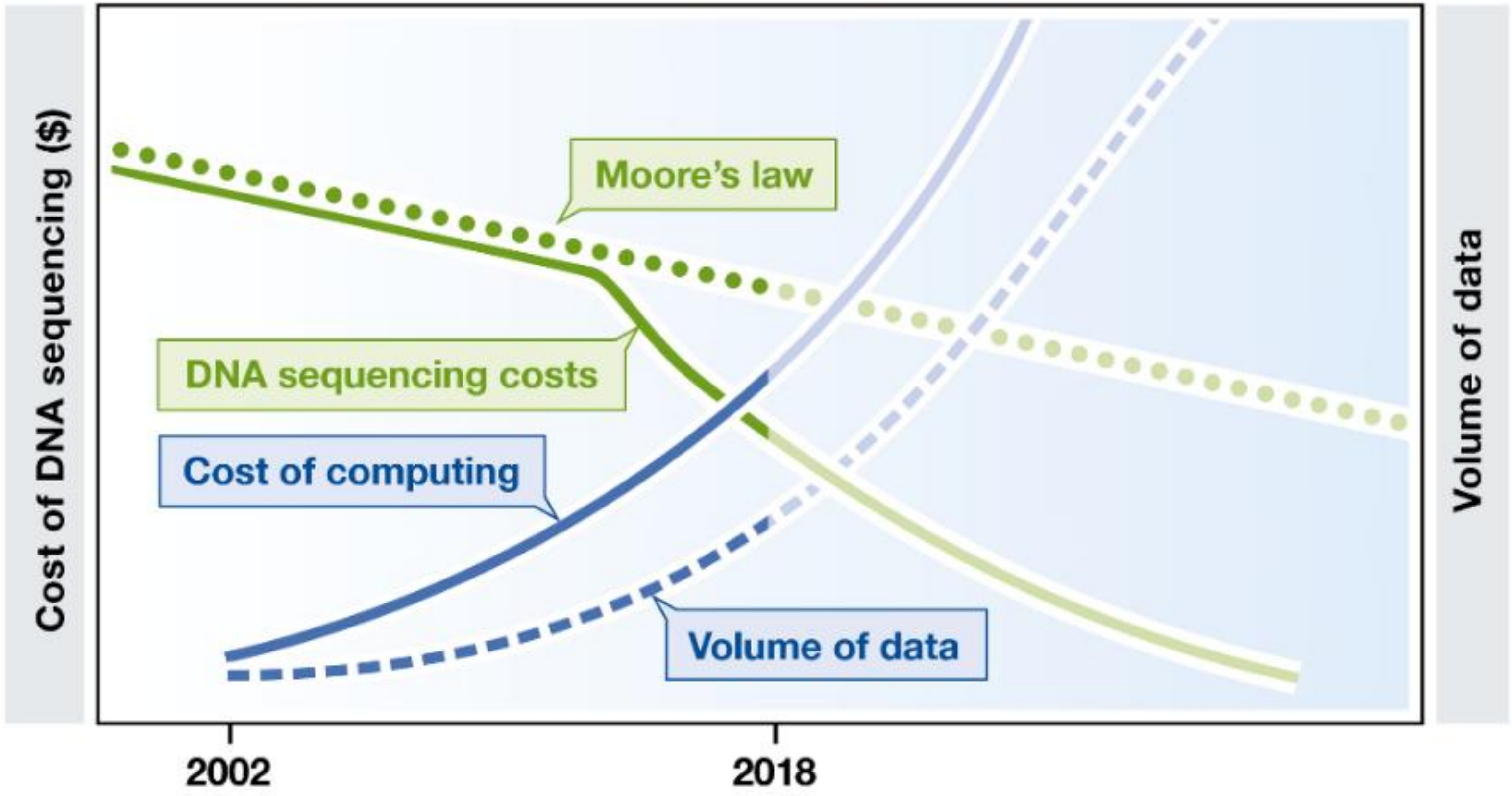
KUZMAN CONSULTING

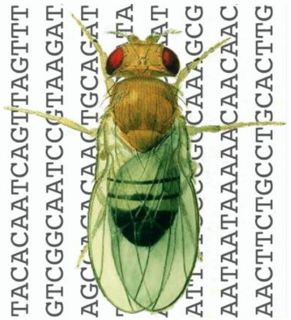
BIOINFORMATICIANS AT YOUR SERVICE


Cost of DNA sequencing ($)
Moore's law
DNA sequencing costs
Cost of computing
Volume of data
2002
2018
Volume of data

NGS baze podataka

Baze bioloških podataka

- podatci u različitom stupnju obrade
- ručno održavane ili “otvorene”



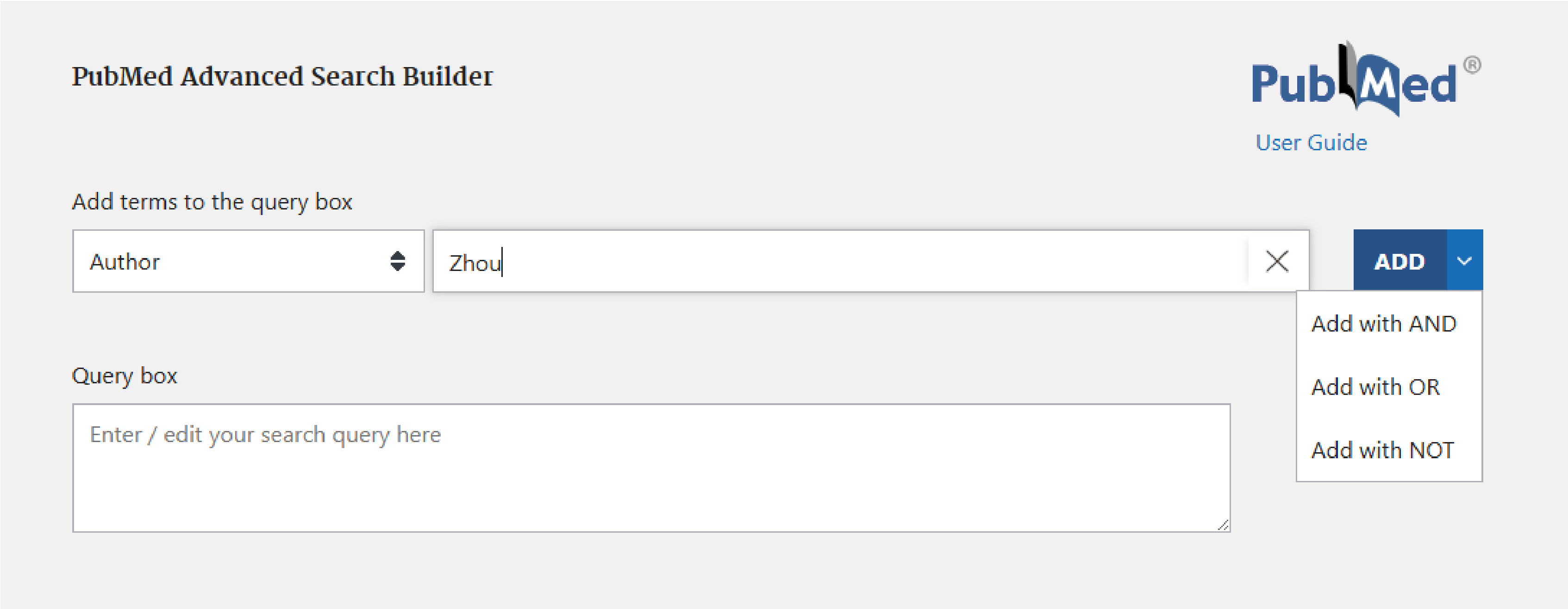
NCBI

- skup međusobno povezanih baza podataka
- dobra početna točka
- sekvence, anotacije, literature, varijante, taksonomija…
- dobra dokumentacija
- napredno pretraživanje

https://www.ncbi.nlm.nih.gov/

PubMed

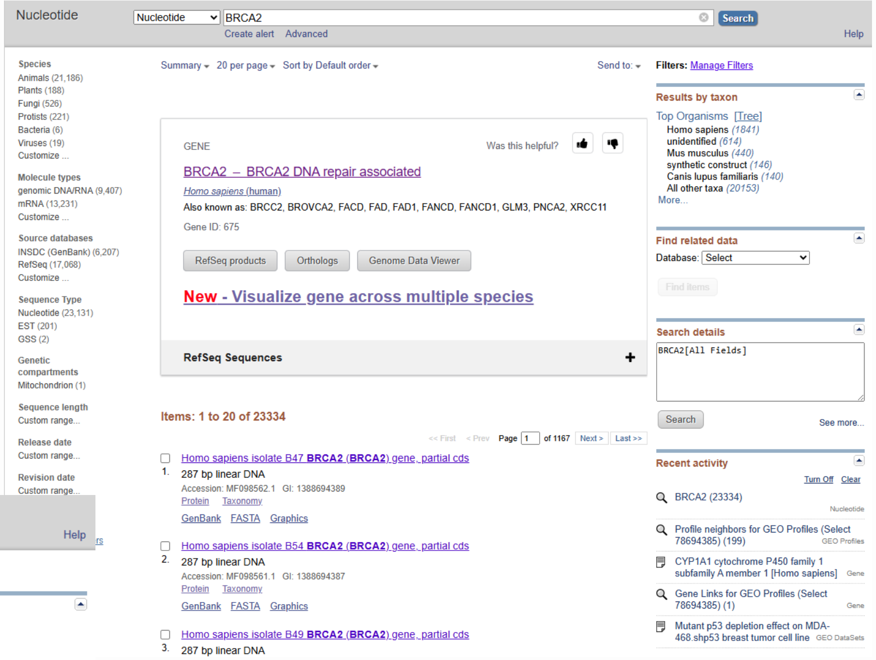
- medicinska i biološka literatura

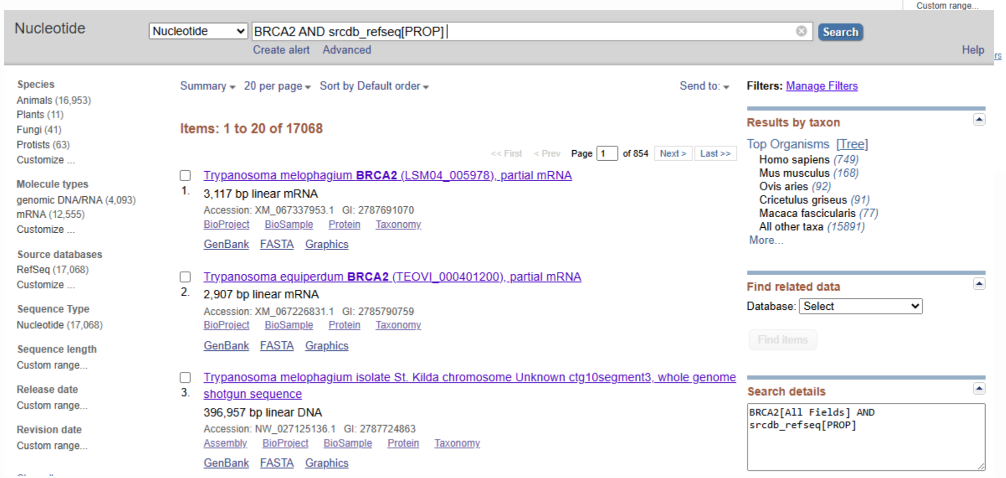


Nucleotide

- GenBank
- RefSeq
- SRA

- Gene
- Genome





Protein

- CDD
- Structure

Citation:

Structure of a meiosis-specific complex central to BRCA2 localization at recombination sites

Pendlebury DF, Zhang J, Agrawal R, Shibuya H, Nandakumar J

Nat Struct Mol Biol (2021) **28** p.671-680

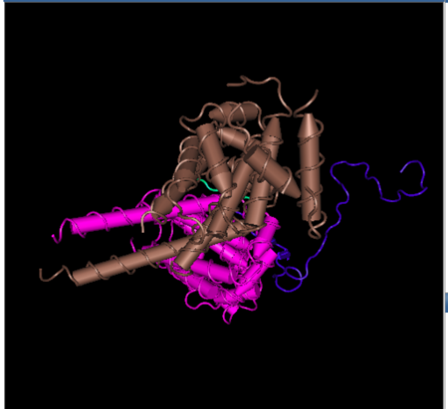
Abstract

Meiotic cells invoke breast cancer susceptibility gene 2 (BRCA2) to repair programmed double-stranded DNA breaks and accomplish homologous recombination. The meiosis-specific protein MEILB2 facilitates BRCA2 recruitment to meiotic recombination sites. Here, we combine crystallography, biochemical analysis and a mouse meiosis model to reveal a robust architecture that ensures... *read more*

PDB ID:	7LDG Download
MMDB ID:	205498
PDB Deposition Date:	2021/1/13
Updated in MMDB:	2024/10
Experimental Method:	x-ray diffraction
Resolution:	2.56 Å
Source Organism:	Homo sapiens
Similar Structures:	VAST+

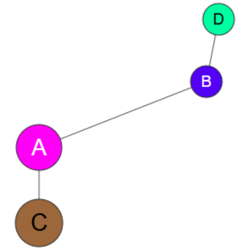
Download sequence data

Molecular Graphic



3D view | full-featured 3D viewer

Interactions



Drag symbols to move
Double click symbols to explore molecules

Download Stucture Data

Download

Format: ASN.1 (Cn3D)
Data Set: alpha-Carbons

Download Cn3D

pfam00634: **BRCA2** (this model, PSSM-Id:459881 is obsolete and has been replaced by 459881) Download alignment

BRCA2 repeat
The alignment covers only the most conserved region of the repeat.

Links

Source:	pfam
Taxonomy:	Eukaryota
PubMed:	1 link
Protein:	Representatives Specific Protein Related Protein Related Structure Architectures
Superfamily:	cl02912

Statistics

Structure

PubMed References

Internal repeats in the BRCA2 protein sequence. *Nat Genet 1996 May ; 13(1):22-3*

Sequence trees and CD hierarchy information associated with this CD can not be displayed.

This model has been replaced with 513619.

Sequence Alignment — include consensus sequence

Format: Hypertext Row Display: up to 10 Color Bits: 2.0 bit Type Selection: the most diverse members

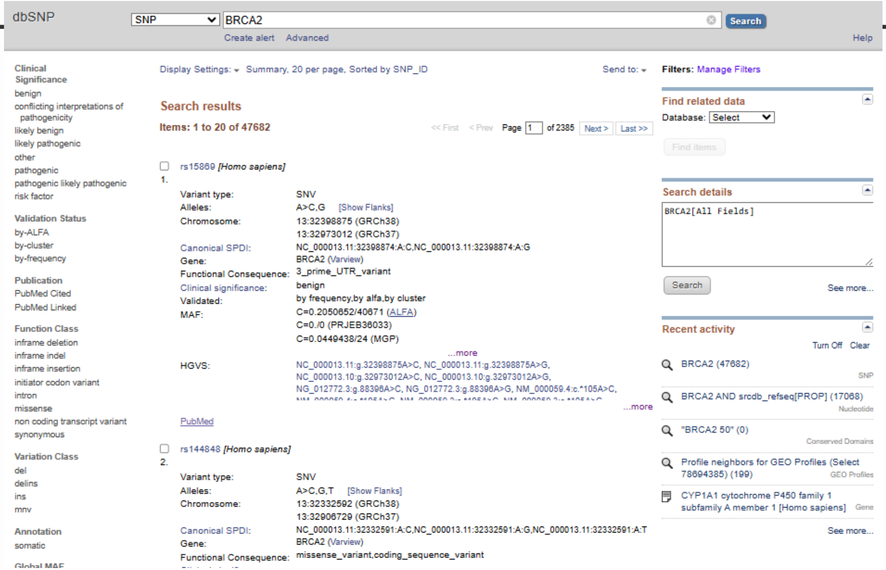
```
XP_001237134  909 FGGGFSTAGGGKISVSKEALERAQKAFAE 937
XP_010727291 2023 PALGFSTASGKQVTISESAYRKAKALLKE 2051
ERM95536      176 GAPLFRSGHGKPVGVAESSIRKAMAVLGE 204
EGV98880     1637 SALTCDTGHGRKTCVSESSLSKDRKWLRD 1665
P97929       1626 SALAYYTEDSRKTCVRESSLSKGRKWLRE 1654 house mouse
XP_019695293 1668 SALSFYTGHGRKISVSQSSILEVKKWLRG 1696
XP_023403030 1650 TALAFYTGRGRQISVSQTSLSEAKKWVRE 1678
XP_012665452 1667 SALAFYTGHGRKISVSQASLFKAKKWLRE 1695
ELK09316     1682 SVLAFYTGHGRKISVRETSLLEARKWLRE 1710
XP_012039996 2058 AFSGFSTASGKHVPVSESALRKVKGMFEE 2086
```

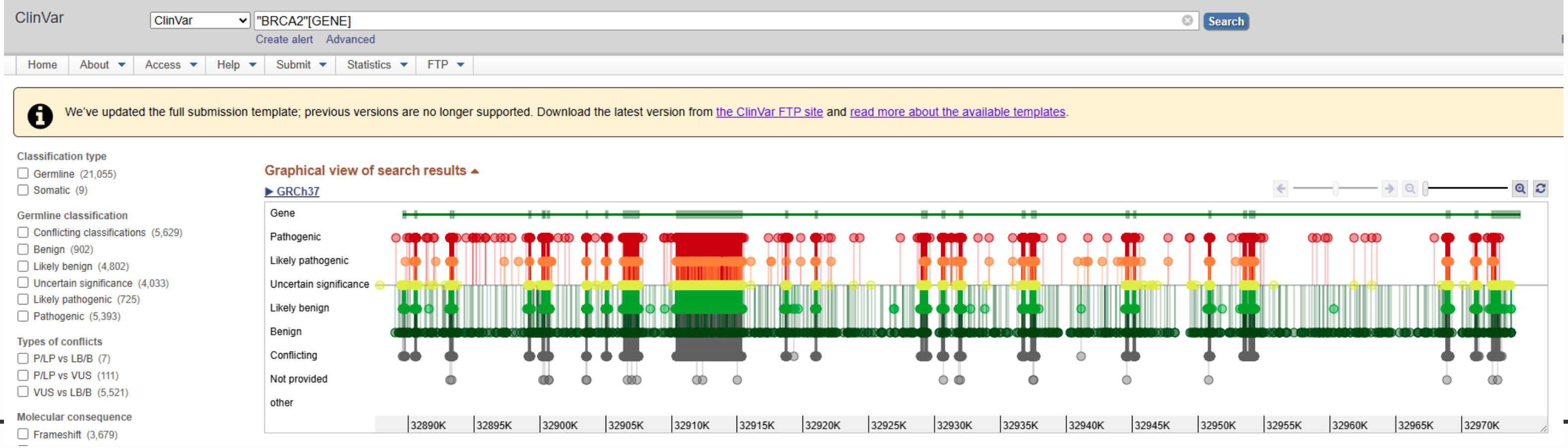
Citing CDD

Jiyao W et al.(2023). *"The conserved domain database in 2023."*, **Nucleic Acids Res. 51(D1)**:D384-D388.

Variation

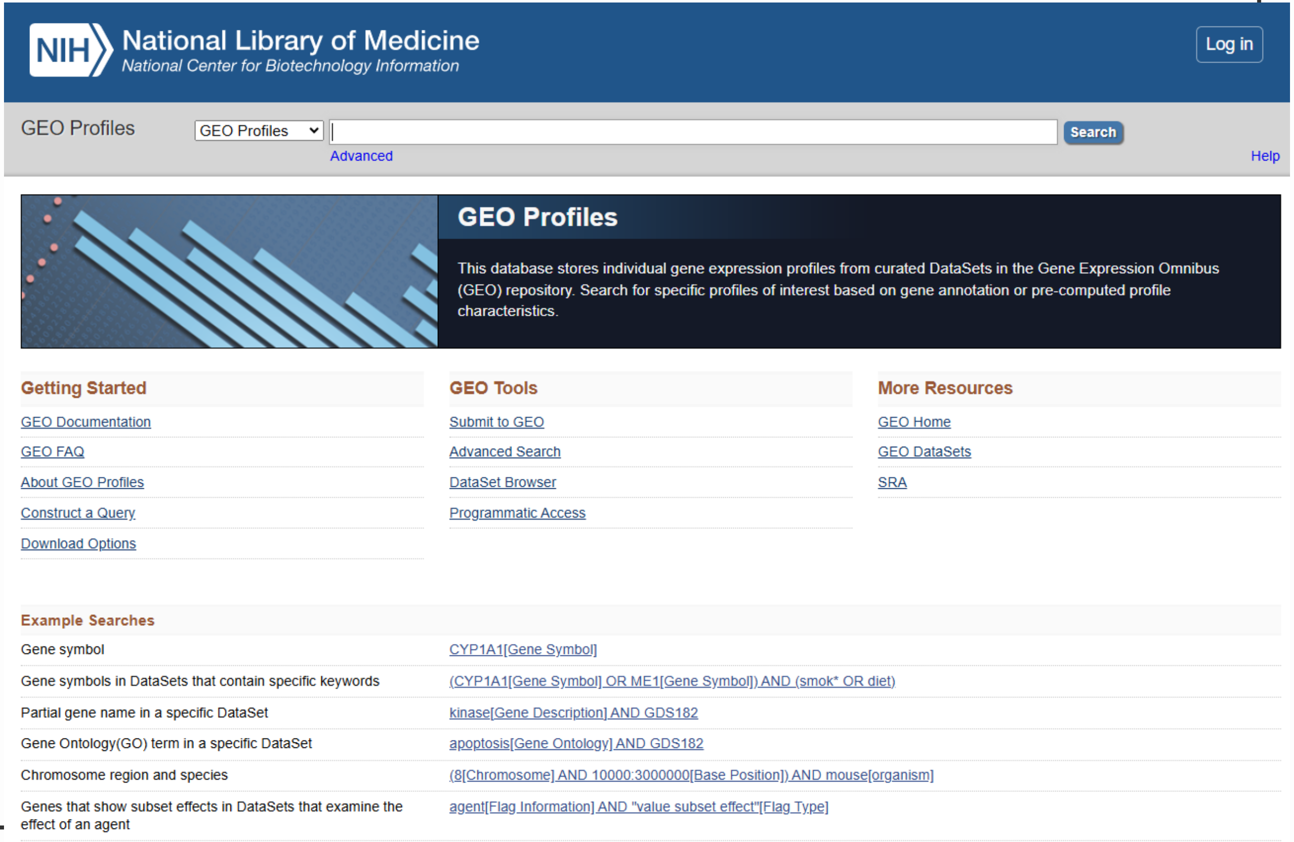
- [dbSNP]
- [ClinVar]





GEO (Gene Expression Omnibus)

- ekspresija
- microarray, RNAseq, rtPCR, SAGE…



Nucleotide

- GenBank
- RefSeq
- SRA

Protein

- CDD
- Structure

Variation

- dbSNP
- ClinVar

Genes & Expression

- BioProject
- GEO
- Online Mendelian Inheritance in Man (OMIM)

Homology

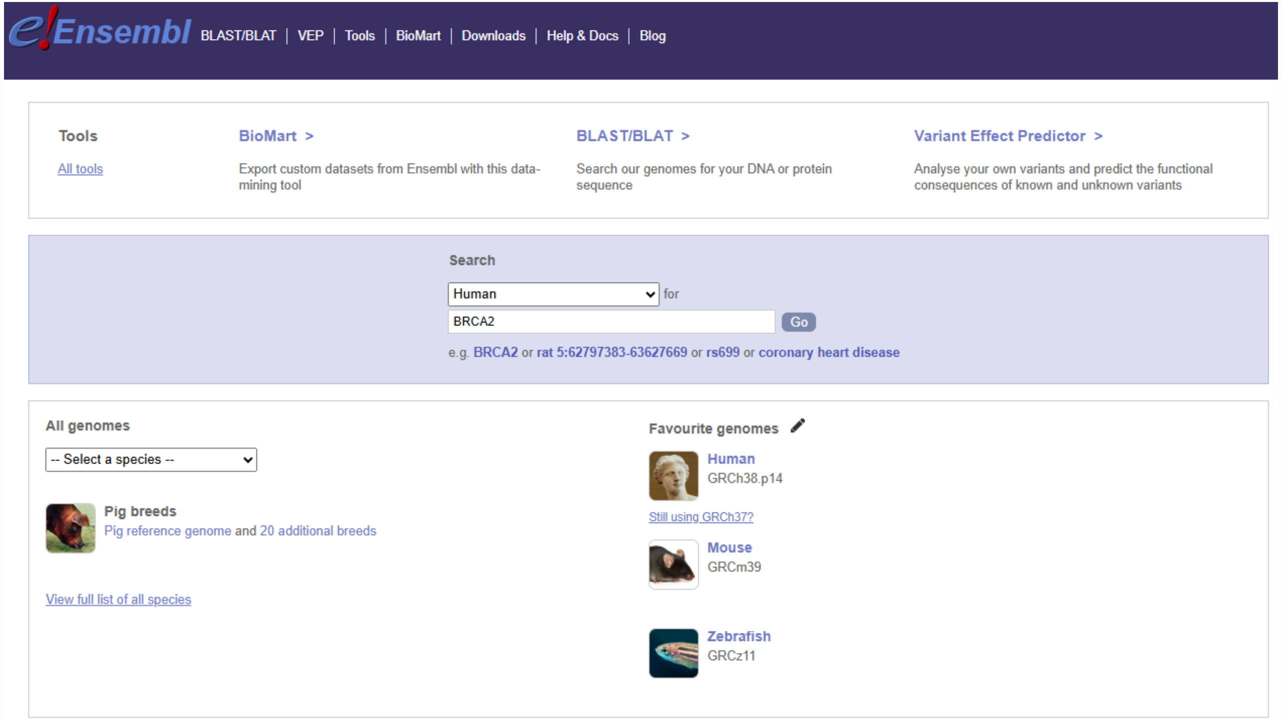
Taxonomy

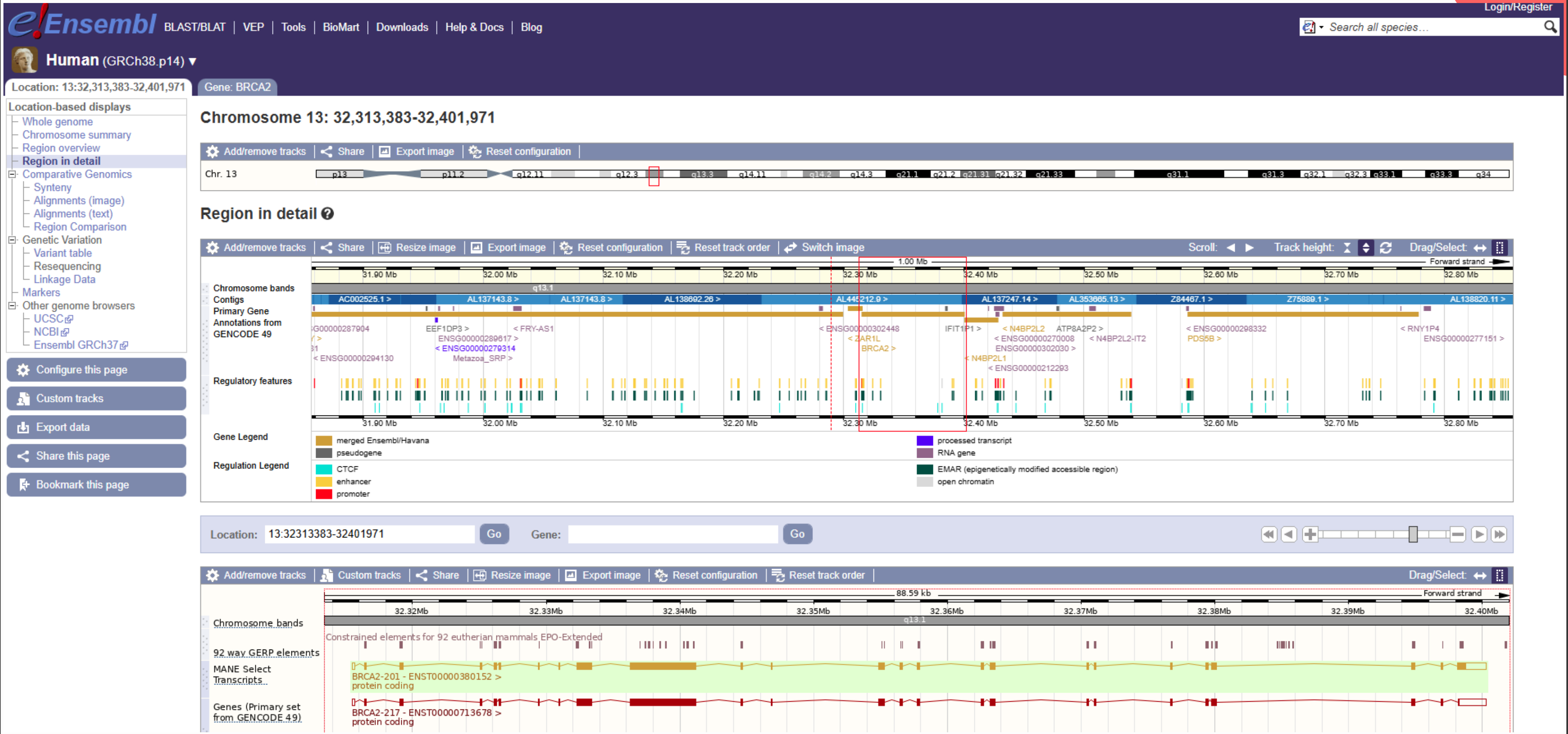
Data & Software

Genomski preglednici

Ensembl

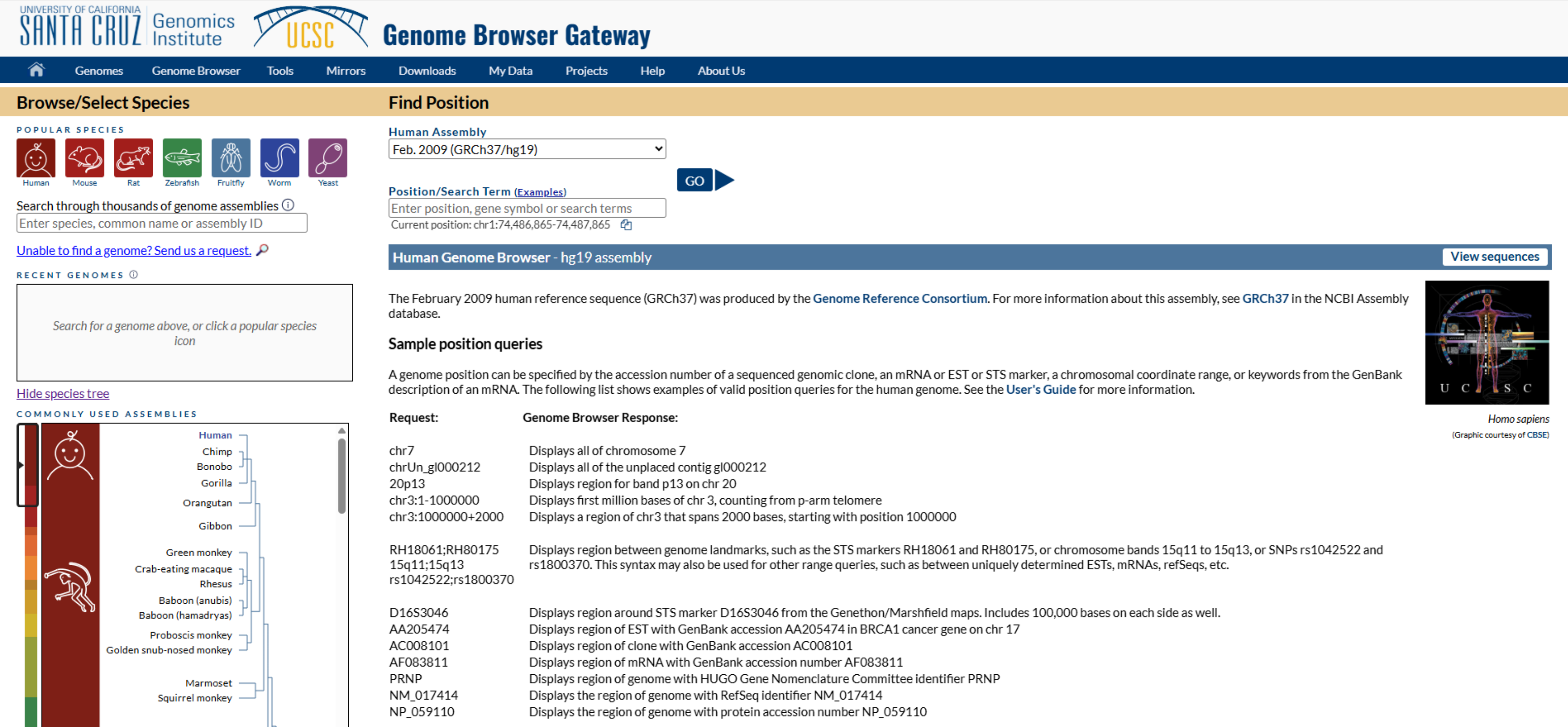
- provjereni podatci
- anotacija
- vizualizacija!



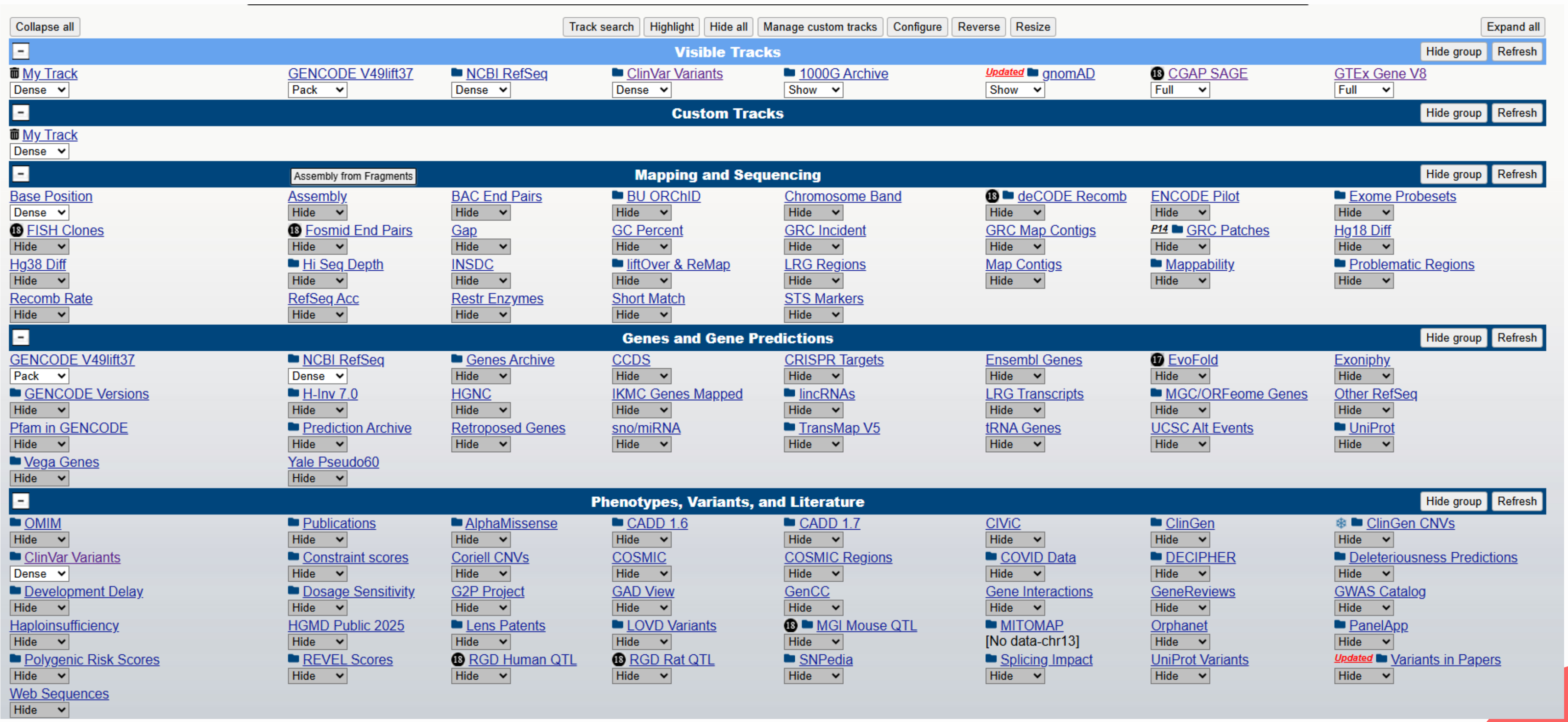


Ensembl
BLAST/BLAT | VEP | Tools | BioMart | Downloads | Help & Docs | Blog
Login/Register
Search all species...
Human (GRCh38.p14)
Location: 13:32,313,383-32,401,971
Gene: BRCA2
Location-based displays
Whole genome
Chromosome summary
Region overview
Region in detail
Comparative Genomics
Synteny
Alignments (image)
Alignments (text)
Region Comparison
Genetic Variation
Variant table
Resequencing
Linkage Data
Markers
Other genome browsers
UCSC
NCBI
Ensembl GRCh37
Configure this page
Custom tracks
Export data
Share this page
Bookmark this page
Chromosome 13: 32,313,383-32,401,971
Add/remove tracks | Share | Export image | Reset configuration
Chr. 13
Region in detail
Add/remove tracks | Share | Resize image | Export image | Reset configuration | Reset track order | Switch image
Scroll:
Track height:
Drag/Select:
1.00 Mb
Forward strand
Chromosome bands
Contigs
Primary Gene Annotations from GENCODE 49
Regulatory features
Gene Legend
merged Ensembl/Havana
pseudogene
processed transcript
RNA gene
Regulation Legend
CTCF
enhancer
promoter
EMAR (epigenetically modified accessible region)
open chromatin
Location: 13:32313383-32401971 Go
Gene: Go
Add/remove tracks | Custom tracks | Share | Resize image | Export image | Reset configuration | Reset track order
Drag/Select:
88.59 kb
Forward strand
Chromosome bands
Constrained elements for 92 eutherian mammals EPO-Extended
92 way GERP elements
MANE Select Transcripts
BRCA2-201 - ENST00000380152 >
protein coding
Genes (Primary set from GENCODE 49)
BRCA2-217 - ENST00000713678 >
protein coding

UCSC Genome Browser



brojne opcije, fleksibilno, mogućnost dodavanja vlastitih podataka!


Collapse all
Track search
Highlight
Hide all
Manage custom tracks
Configure
Reverse
Resize
Expand all
Visible Tracks
Hide group
Refresh
My Track
Dense
GENCODE V49lift37
Pack
NCBI RefSeq
Dense
ClinVar Variants
Dense
1000G Archive
Show
Updated
gnomAD
Show
CGAP SAGE
Full
GTEx Gene V8
Full
Custom Tracks
Hide group
Refresh
My Track
Dense
Assembly from Fragments
Mapping and Sequencing
Hide group
Refresh
Base Position
Dense
Assembly
Hide
BAC End Pairs
Hide
BU ORChID
Hide
Chromosome Band
Hide
deCODE Recomb
Hide
ENCODE Pilot
Hide
Exome Probesets
Hide
FISH Clones
Hide
Fosmid End Pairs
Hide
Gap
Hide
GC Percent
Hide
GRC Incident
Hide
GRC Map Contigs
Hide
P14
GRC Patches
Hide
Hg18 Diff
Hide
Hg38 Diff
Hide
Hi Seq Depth
Hide
INSDC
Hide
liftOver & ReMap
Hide
LRG Regions
Hide
Map Contigs
Hide
Mappability
Hide
Problematic Regions
Hide
Recomb Rate
Hide
RefSeq Acc
Hide
Restr Enzymes
Hide
Short Match
Hide
STS Markers
Hide
Genes and Gene Predictions
Hide group
Refresh
GENCODE V49lift37
Pack
NCBI RefSeq
Dense
Genes Archive
Hide
CCDS
Hide
CRISPR Targets
Hide
Ensembl Genes
Hide
EvoFold
Hide
Exoniphy
Hide
GENCODE Versions
Hide
H-Inv 7.0
Hide
HGNC
Hide
IKMC Genes Mapped
Hide
lincRNAs
Hide
LRG Transcripts
Hide
MGC/ORFeome Genes
Hide
Other RefSeq
Hide
Pfam in GENCODE
Hide
Prediction Archive
Hide
Retroposed Genes
Hide
sno/miRNA
Hide
TransMap V5
Hide
tRNA Genes
Hide
UCSC Alt Events
Hide
UniProt
Hide
Vega Genes
Hide
Yale Pseudo60
Hide
Phenotypes, Variants, and Literature
Hide group
Refresh
OMIM
Hide
Publications
Hide
AlphaMissense
Hide
CADD 1.6
Hide
CADD 1.7
Hide
CIViC
Hide
ClinGen
Hide
ClinGen CNVs
Hide
ClinVar Variants
Dense
Constraint scores
Hide
Coriell CNVs
Hide
COSMIC
Hide
COSMIC Regions
Hide
COVID Data
Hide
DECIPHER
Hide
Deleteriousness Predictions
Hide
Development Delay
Hide
Dosage Sensitivity
Hide
G2P Project
Hide
GAD View
Hide
GenCC
Hide
Gene Interactions
Hide
GeneReviews
Hide
GWAS Catalog
Hide
Haploinsufficiency
Hide
HGMD Public 2025
Hide
Lens Patents
Hide
LOVD Variants
Hide
MGI Mouse QTL
Hide
MITOMAP
[No data-chr13]
Orphanet
Hide
PanelApp
Hide
Polygenic Risk Scores
Hide
REVEL Scores
Hide
RGD Human QTL
Hide
RGD Rat QTL
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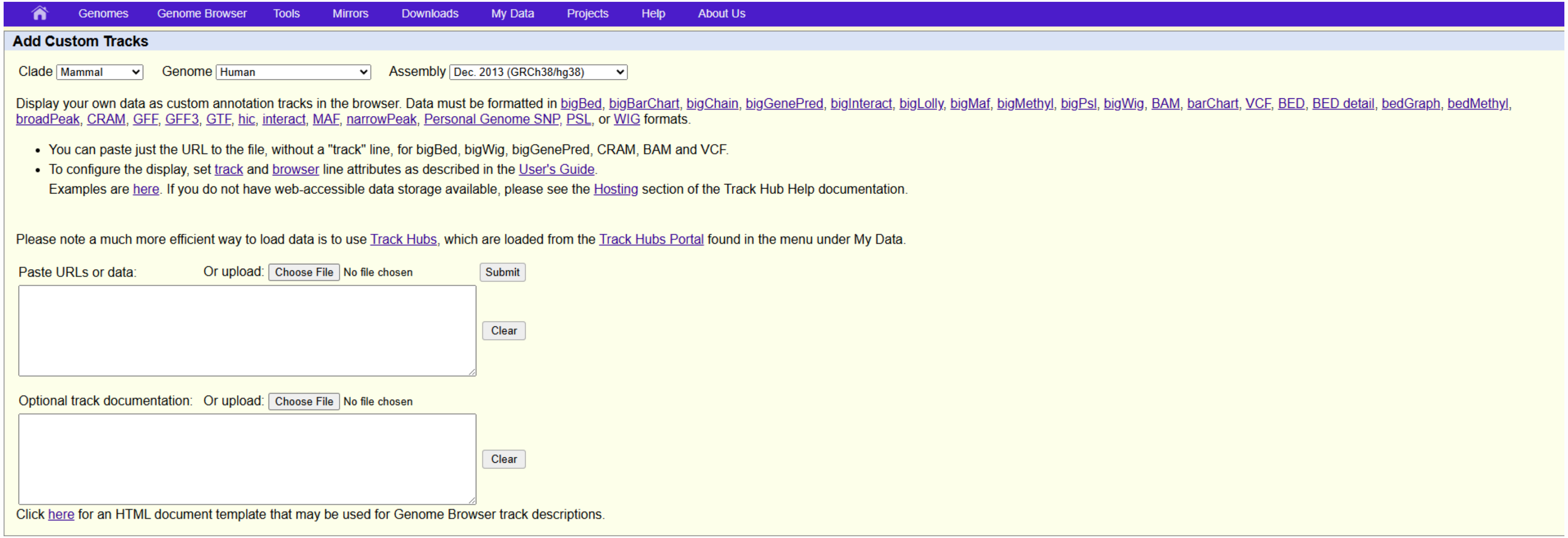
SNPedia
Hide
Splicing Impact
Hide
UniProt Variants
Hide
Updated
Variants in Papers
Hide
Web Sequences
Hide


Genomes
Genome Browser
Tools
Mirrors
Downloads
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Projects
Help
About Us
Add Custom Tracks
Clade Mammal
Genome Human
Assembly Dec. 2013 (GRCh38/hg38)
Display your own data as custom annotation tracks in the browser. Data must be formatted in bigBed, bigBarChart, bigChain, bigGenePred, bigInteract, bigLolly, bigMaf, bigMethyl, bigPsl, bigWig, BAM, barChart, VCF, BED, BED detail, bedGraph, bedMethyl, broadPeak, CRAM, GFF, GFF3, GTF, hic, interact, MAF, narrowPeak, Personal Genome SNP, PSL, or WIG formats.
You can paste just the URL to the file, without a "track" line, for bigBed, bigWig, bigGenePred, CRAM, BAM and VCF.
To configure the display, set track and browser line attributes as described in the User's Guide.
Examples are here. If you do not have web-accessible data storage available, please see the Hosting section of the Track Hub Help documentation.
Please note a much more efficient way to load data is to use Track Hubs, which are loaded from the Track Hubs Portal found in the menu under My Data.
Paste URLs or data:
Or upload: Choose File No file chosen
Submit
Clear
Optional track documentation:
Or upload: Choose File No file chosen
Clear
Click here for an HTML document template that may be used for Genome Browser track descriptions.

tutoriali

Tipovi podataka

Genomski podatci

- biološki podatci su višedimenzionalni - izazov
- “sirove” sekvence, sekvence s meta-podatcima, sekvence u međusobnom odnosu, podatci o funkciji, jačina signala....
- plus meta-podatci o uzorku: priprema, histologija, lokacija, dob, spol....

Genomski podatci

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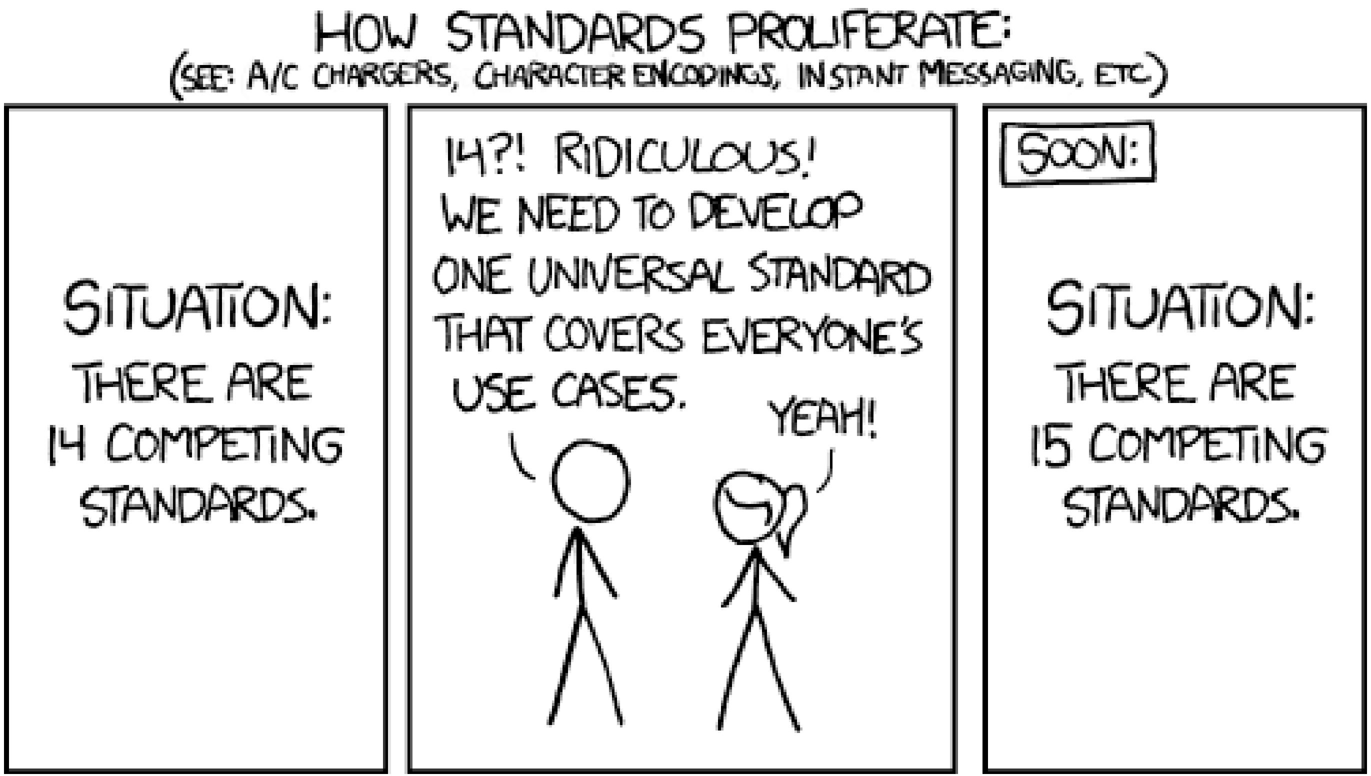
- plus meta-podatci o uzorku: priprema, histologija, lokacija, dob, spol....
- neefikasno – većinom tekstualni formati
- nestandardizirano


HOW STANDARDS PROLIFERATE:
(SEE: A/C CHARGERS, CHARACTER ENCODINGS, INSTANT MESSAGING, ETC)
SITUATION:
THERE ARE
14 COMPETING
STANDARDS.
14?! RIDICULOUS!
WE NEED TO DEVELOP
ONE UNIVERSAL STANDARD
THAT COVERS EVERYONE'S
USE CASES.
YEAH!
SOON:
SITUATION:
THERE ARE
15 COMPETING
STANDARDS.

FASTA i FASTQ

- osnovni format za spremanje (nukleotidnih) sekvenci

```
>sequenceID-001 description
AAGTAGGAATAATATCTTATCATTATAGATAAAAACCTTCTGAATTTGCTTAGTGTGTAT
ACGACTAGACATATATCAGCTCGCCGATTATTTGGATTATTCCCTG
>sequenceID-002 description
CAGTAAAGAGTGGATGTAAGAACCGTCCGATCTACCAGATGTGATAGAGGTTGCCAGTAC
AAAAATTGCATAATAATTGATTAATCCTTTAATATTGTTTAGAATATATCCGTCAGATAA
TCCTAAAAATAACGATATGATGGCGGAAATCGTC
>sequenceID-003 description
CTTCAATTACCCTGCTGACGCGAGATACCTTATGCATCGAAGGTAAAGCGATGAATTTAT
CCAAGGTTTTAATTTG
```

FASTA i FASTQ

- osnovni format za spremanje (nukleotidnih) sekvenci
- FASTQ sadrži i podatke o kvaliteti očitanja

```
@A0216:173:HJNFKDSX:3:1101:2745:1016 1:N:0:TTACGCAC+AGATGGTC
GGGAGACGAGCGTCACGTTCATGAGCGCCTCCTCGACCTCGTCGACGGAGTCGAAATTACGCGCCACCCACAGCGGAAATTCGAACGTGGCGACGTTTTCC
+
BBBFBFFFFFFFBF<FBB0B<BF<FFFFFIIII'<<FF7BF<'7<BBBBB<<BBBBBBBBBBBBBBBBBBBBB<7BB<<B<BFB<BBBB77<BB<BBBBB
@A0216:173:HJNFKDSX:3:1101:3884:1016 1:N:0:TTACGCAC+AGATGGTC
GCTTCGCCTTGCCGCCCGCACCAAACACCGTGTCATAGTGGTAGCCGCGGAGTAACCAAGATGGTCTCCCCATATGAGAAACTCCAGTCGAGATTACGG
+
BBBFFFFFFFFFFFFFIBFFIIIFFIIBBF0BFFF0BF<BF<BBFFBFFB<BBFBBBF0<BBFFFBBBFBBFFFBBBF<<BBBBFFFFB<BBFFFBBBB07
@A0216:173:HJNFKDSX:3:1101:4010:1016 1:N:0:TTACGCAC+AGATGGTC
TATTTTTATGCATTTGCTTAATTTTAATAGCTGCAAAAGTTAATAAAATAAAACCAATTGCTACAAAAAAAGAAGTGCAGCTTCAAGAAGATGTAAAACTG
+
<<<<<0<00000<BB7BBBB000BB0<7BB000B<0BB'<<<'0'0'<0<00'07BB'77''07<7''0000'7<<0'0<<''7<'7'77'000'77'''7
```

FASTA i FASTQ

```
(33):  !"#$%&'()*+,-./0123456789:;<=>?@ABCDEFGHI
(64):  @ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefgh
```

$$Q = -10 \log_{10} P \quad \longrightarrow \quad P = 10^{\frac{-Q}{10}}$$

Phred Quality Score	Probability of incorrect base call	Base call accuracy
10	1 in 10	90%
20	1 in 100	99%
30	1 in 1000	99.9%
40	1 in 10000	99.99%
50	1 in 100000	99.999%

SAM i BAM

- spremanje mapiranih podataka
- zaglavlje + 11 obaveznih kolona

Col	Field	Type	Regexp/Range	Brief description
1	QNAME	String	`[!-?A-~]{1,254}`	Query template NAME
2	FLAG	Int	$[0, 2^{16}-1]$	bitwise FLAG
3	RNAME	String	`\*|[:rname:^*=][:rname:]*`	Reference sequence NAME[12]
4	POS	Int	$[0, 2^{31}-1]$	1-based leftmost mapping POSition
5	MAPQ	Int	$[0, 2^{8}-1]$	MAPping Quality
6	CIGAR	String	`\*|([0-9]+[MIDNSHP=X])+`	CIGAR string
7	RNEXT	String	`\*|=|[:rname:^*=][:rname:]*`	Reference name of the mate/next read
8	PNEXT	Int	$[0, 2^{31}-1]$	Position of the mate/next read
9	TLEN	Int	$[-2^{31}+1, 2^{31}-1]$	observed Template LENgth
10	SEQ	String	`\*|[A-Za-z=.]+`	segment SEQuence
11	QUAL	String	`[!-~]+`	ASCII of Phred-scaled base QUALity+33

SAM i BAM

- spremanje mapiranih podataka
- zaglavlje + 11 obaveznih kolona – CIGAR string opisuje poravnanje

```
@HD VN:1.6 SO:coordinate
@SQ SN:ref LN:45
r001   99 ref  7 30 8M2I4M1D3M = 37  39 TTAGATAAAGGATACTG *
r002    0 ref  9 30 3S6M1P1I4M *  0   0 AAAAGATAAGGATA    *
r003    0 ref  9 30 5S6M       *  0   0 GCCTAAGCTAA       * SA:Z:ref,29,-,6H5M,17,0;
r004    0 ref 16 30 6M14N5M    *  0   0 ATAGCTTCAGC       *
r003 2064 ref 29 17 6H5M       *  0   0 TAGGC             * SA:Z:ref,9,+,5S6M,30,1;
r001  147 ref 37 30 9M         =  7 -39 CAGCGGCAT         * NM:i:1
```

SAM i BAM

Header section

```
@HD	VN:1.3	SO:coordinate
@SQ	SN:conticA	LN:443
@SQ	SN:contigB	LN:1493
@SQ	SN:contigC	LN:328
```

Tab-delimited read alignment information lines

```
readID43GYAX15:7:1:1202:19894/1    256    contig43    613960    1    65M    *    0    0
CCAGCGCGAACGAAATCCGCATGCGTCTGGTCGTTGCACGGAACGGCGGCGGTGTGATGCACGGC    EDDEEDEE=EE?DE??DDDBADEBEFFFDBEFFEBCBC=?BEEEE@=:?::?7?:8-
6?7?@??#    AS:i:0    XS:i:0   XN:i:0   XM:i:0   XO:i:0   XG:i:0   NM:i:0  MD:Z:65  YT:Z:UU
readID43GYAX15:7:1:1202:19894/1    272    contig32    21001    1    65M    *    0    0
GCCGGACGTCACACGGCCGCCGGGCCGGTCTACGACCAGACGCATGCGGATTTCGTTAGAGCCGG    #??@?7?6-8:?7?::?:=@EEEEB?=CBCBEFFEBDFFFEBEDABDDD??ED?
EE=EEDEEDDE    AS:i:-5    XS:i:0    XN:i:0   XM:i:1    XO:i:0    XG:i:0    NM:i:1    MD:Z:42T22    YT:Z:UU
readID43GYAX15:7:1:1202:19894/1    256    contig87    540849    1    65M    *    0    0
CCTGCACGAACGAAATCCGCATGCGTCTGGTCGTTGTACGGAACGGCGGTTGTGTGACGAACGGC    EDDEEDEE=EE?DE??DDDBADEBEFFFDBEFFEBCBC=?BEEEE@=:?::?7?:8-
6?7?@??# AS:i:0  XS:i:0 XN:i:0  XM:i:0 XO:i:0  XG:i:0  NM:i:0    MD:Z:65    YT:Z:UU
```

BED

- jednostavan tablični format

```
track name="AML v2 amplicons" description="Tapestri Designer: AML v2 panel: amplicon
regions" type=bed db=hg19 color=77,175,74 priority=2
chr1    115256487       115256723       AML_v2_NRAS_115256512
chr1    115258609       115258825       AML_v2_NRAS_115258635
chr2    25457143        25457372        AML_v2_DNMT3A_25457166
chr2    25458518        25458763        AML_v2_DNMT3A_25458540
chr2    25459793        25460046        AML_v2_DNMT3A_25459813
chr2    25461880        25462137        AML_v2_DNMT3A_25461902
chr2    25463106        25463346        AML_v2_DNMT3A_25463127
chr2    25463493        25463717        AML_v2_DNMT3A_25463515
chr2    25464421        25464618        AML_v2_DNMT3A_25464443
chr2    25466621        25466871        AML_v2_DNMT3A_25466642
chr2    25467012        25467220        AML_v2_DNMT3A_25467033
chr2    25467371        25467631        AML_v2_DNMT3A_25467391
chr2    25468110        25468332        AML_v2_DNMT3A_25468130
chr2    25469006        25469231        AML_v2_DNMT3A_25469026
```

GTF
GFF(3)

Gene Transfer

file Format

General

Feature Format

```
##gff-version 3.2.1
##sequence-region ctg123 1 1497228
ctg123 . gene            1000  9000  .  +  .  ID=gene00001;Name=EDEN
ctg123 . TF_binding_site 1000  1012  .  +  .  ID=tfbs00001;Parent=gene00001
ctg123 . mRNA            1050  9000  .  +  .  ID=mRNA00001;Parent=gene00001;Name=EDEN.1
ctg123 . mRNA            1050  9000  .  +  .  ID=mRNA00002;Parent=gene00001;Name=EDEN.2
ctg123 . mRNA            1300  9000  .  +  .  ID=mRNA00003;Parent=gene00001;Name=EDEN.3
ctg123 . exon            1300  1500  .  +  .  ID=exon00001;Parent=mRNA00003
ctg123 . exon            1050  1500  .  +  .  ID=exon00002;Parent=mRNA00001,mRNA00002
ctg123 . exon            3000  3902  .  +  .  ID=exon00003;Parent=mRNA00001,mRNA00003
ctg123 . exon            5000  5500  .  +  .  ID=exon00004;Parent=mRNA00001,mRNA00002,mRNA00003
ctg123 . exon            7000  9000  .  +  .  ID=exon00005;Parent=mRNA00001,mRNA00002,mRNA00003
ctg123 . CDS             1201  1500  .  +  0  ID=cds00001;Parent=mRNA00001;Name=edenprotein.1
ctg123 . CDS             3000  3902  .  +  0  ID=cds00001;Parent=mRNA00001;Name=edenprotein.1
ctg123 . CDS             5000  5500  .  +  0  ID=cds00001;Parent=mRNA00001;Name=edenprotein.1
ctg123 . CDS             7000  7600  .  +  0  ID=cds00001;Parent=mRNA00001;Name=edenprotein.1
ctg123 . CDS             1201  1500  .  +  0  ID=cds00002;Parent=mRNA00002;Name=edenprotein.2
ctg123 . CDS             5000  5500  .  +  0  ID=cds00002;Parent=mRNA00002;Name=edenprotein.2
ctg123 . CDS             7000  7600  .  +  0  ID=cds00002;Parent=mRNA00002;Name=edenprotein.2
ctg123 . CDS             3301  3902  .  +  0  ID=cds00003;Parent=mRNA00003;Name=edenprotein.3
ctg123 . CDS             5000  5500  .  +  1  ID=cds00003;Parent=mRNA00003;Name=edenprotein.3
ctg123 . CDS             7000  7600  .  +  1  ID=cds00003;Parent=mRNA00003;Name=edenprotein.3
ctg123 . CDS             3391  3902  .  +  0  ID=cds00004;Parent=mRNA00003;Name=edenprotein.4
ctg123 . CDS             5000  5500  .  +  1  ID=cds00004;Parent=mRNA00003;Name=edenprotein.4
ctg123 . CDS             7000  7600  .  +  1  ID=cds00004;Parent=mRNA00003;Name=edenprotein.4
```

VCF

- Variant Call Format

```
##fileformat=VCFv4.3
##fileDate=20090805
##source=myImputationProgramV3.1
##reference=file:///seq/references/1000GenomesPilot-NCBI36.fasta
##contig=<ID=20,length=62435964,assembly=B36,md5=f126cdf8a6e0c7f379d618ff66beb2da,species="Homo sapiens",taxonomy=x>
##phasing=partial
##INFO=<ID=NS,Number=1,Type=Integer,Description="Number of Samples With Data">
##INFO=<ID=DP,Number=1,Type=Integer,Description="Total Depth">
##INFO=<ID=AF,Number=A,Type=Float,Description="Allele Frequency">
##INFO=<ID=AA,Number=1,Type=String,Description="Ancestral Allele">
##INFO=<ID=DB,Number=0,Type=Flag,Description="dbSNP membership, build 129">
##INFO=<ID=H2,Number=0,Type=Flag,Description="HapMap2 membership">
##FILTER=<ID=q10,Description="Quality below 10">
##FILTER=<ID=s50,Description="Less than 50% of samples have data">
##FORMAT=<ID=GT,Number=1,Type=String,Description="Genotype">
##FORMAT=<ID=GQ,Number=1,Type=Integer,Description="Genotype Quality">
##FORMAT=<ID=DP,Number=1,Type=Integer,Description="Read Depth">
##FORMAT=<ID=HQ,Number=2,Type=Integer,Description="Haplotype Quality">
#CHROM POS     ID        REF    ALT     QUAL FILTER INFO                              FORMAT      NA00001        NA00002        NA00003
20     14370   rs6054257 G      A       29   PASS   NS=3;DP=14;AF=0.5;DB;H2           GT:GQ:DP:HQ 0|0:48:1:51,51 1|0:48:8:51,51 1/1:43:5:.,.
20     17330   .         T      A       3    q10    NS=3;DP=11;AF=0.017               GT:GQ:DP:HQ 0|0:49:3:58,50 0|1:3:5:65,3   0/0:41:3
20     1110696 rs6040355 A      G,T     67   PASS   NS=2;DP=10;AF=0.333,0.667;AA=T;DB GT:GQ:DP:HQ 1|2:21:6:23,27 2|1:2:0:18,2   2/2:35:4
20     1230237 .         T      .       47   PASS   NS=3;DP=13;AA=T                   GT:GQ:DP:HQ 0|0:54:7:56,60 0|0:48:4:51,51 0/0:61:2
20     1234567 microsat1 GTC    G,GTCT  50   PASS   NS=3;DP=9;AA=G                    GT:GQ:DP    0/1:35:4       0/2:17:2       1/1:40:3
```

Specifikacije formata:

- https://www.ensembl.org/info/website/upload/index.html
- https://genome.ucsc.edu/FAQ/FAQformat.html
- https://genome.ucsc.edu/ENCODE/fileFormats.html

Bioinformatički alati

Alati u bioinformatici

lh3/bwa

Burrow-Wheeler Aligner for short-read alignment
(see minimap2 for long-read alignment)

Samtools

R

bedtools

strelka Public

biopython

VeP

MultiQC

picard Public

Bioconductor

CLUSTAL

seqtk

Nucleotide BLAST
nucleotide ▸ nucleotide

FastQC

nextflow

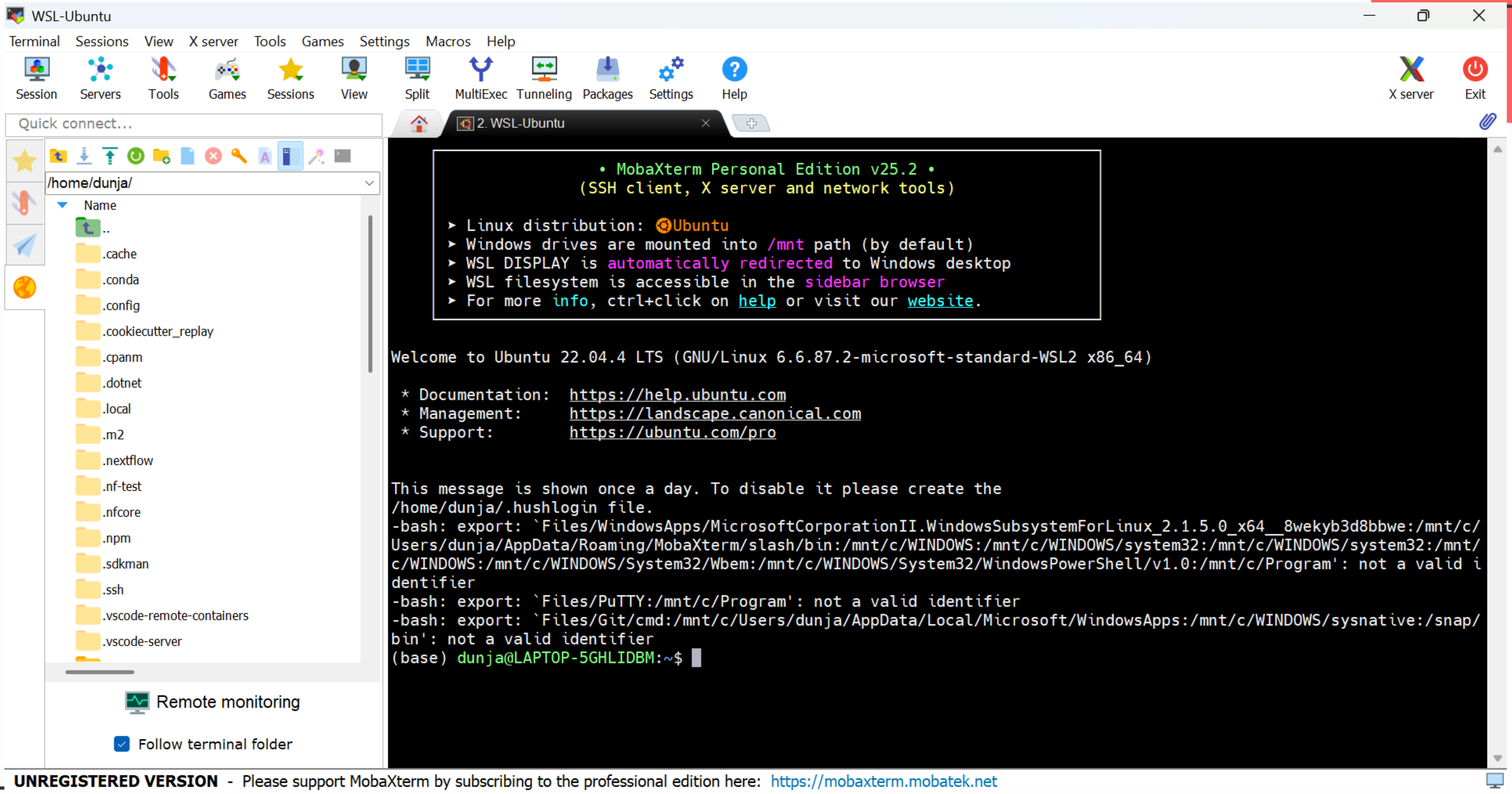
gatk

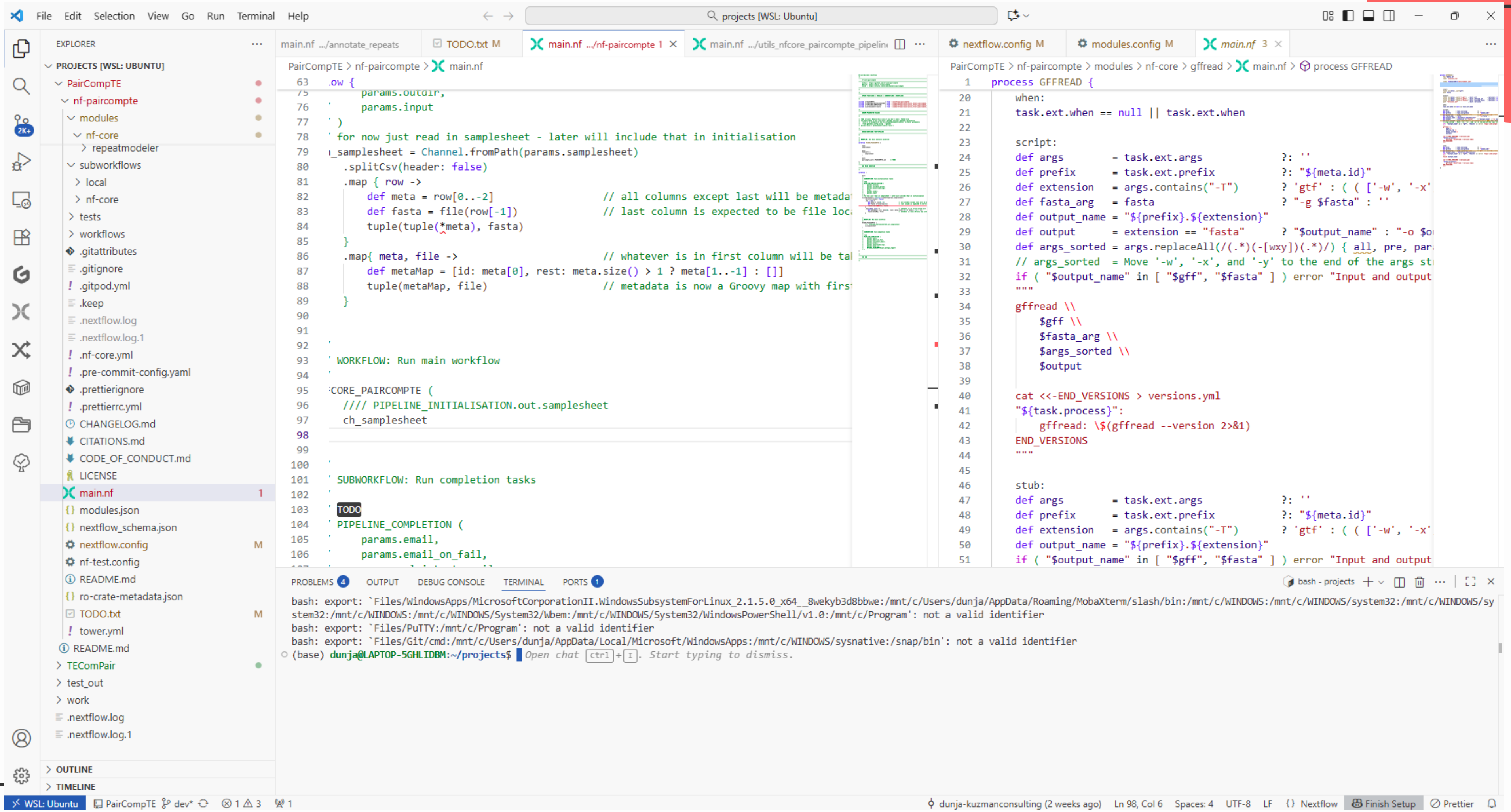
Alati u bioinformatici - prednosti

- velik broj alata za skoro sve tipove analiza
- većina je open-source
- forumi i zajednice: SEQanswers, Biostars, GitHub issues

Alati u bioinformatici - mane

- velik broj alata za skoro sve tipove analiza
- većina je open-source

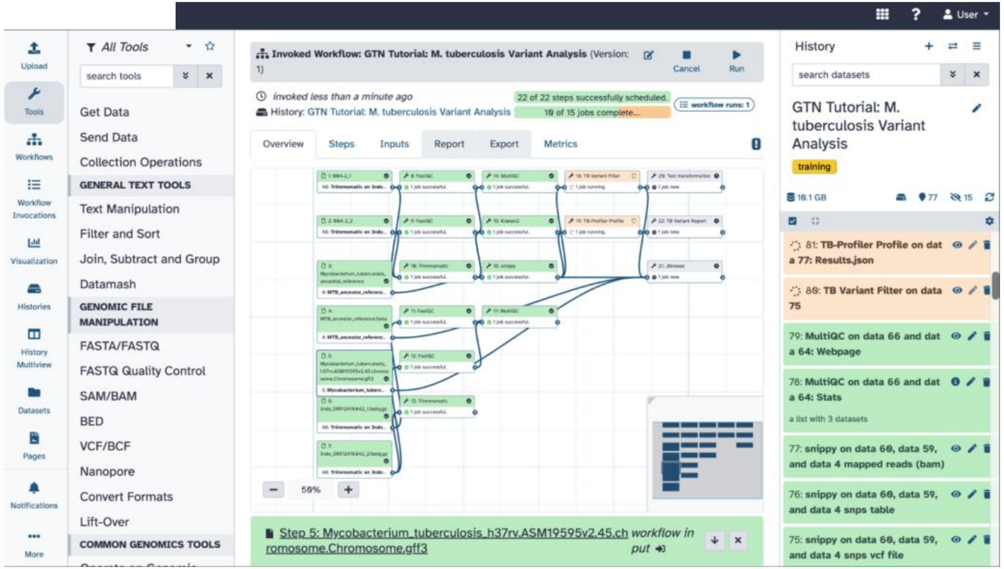

WSL-Ubuntu
Terminal Sessions View X server Tools Games Settings Macros Help
Session Servers Tools Games Sessions View Split MultiExec Tunneling Packages Settings Help X server Exit
Quick connect...
2. WSL-Ubuntu
/home/dunja/
Name
..
.cache
.conda
.config
.cookiecutter_replay
.cpanm
.dotnet
.local
.m2
.nextflow
.nf-test
.nfcore
.npm
.sdkman
.ssh
.vscode-remote-containers
.vscode-server
Remote monitoring
Follow terminal folder
• MobaXterm Personal Edition v25.2 •
(SSH client, X server and network tools)
► Linux distribution: Ubuntu
► Windows drives are mounted into /mnt path (by default)
► WSL DISPLAY is automatically redirected to Windows desktop
► WSL filesystem is accessible in the sidebar browser
► For more info, ctrl+click on help or visit our website.
Welcome to Ubuntu 22.04.4 LTS (GNU/Linux 6.6.87.2-microsoft-standard-WSL2 x86_64)
* Documentation: https://help.ubuntu.com
* Management: https://landscape.canonical.com
* Support: https://ubuntu.com/pro
This message is shown once a day. To disable it please create the
/home/dunja/.hushlogin file.
-bash: export: `Files/WindowsApps/MicrosoftCorporationII.WindowsSubsystemForLinux_2.1.5.0_x64__8wekyb3d8bbwe:/mnt/c/Users/dunja/AppData/Roaming/MobaXterm/slash/bin:/mnt/c/WINDOWS:/mnt/c/WINDOWS/system32:/mnt/c/WINDOWS/system32:/mnt/c/WINDOWS:/mnt/c/WINDOWS/System32/Wbem:/mnt/c/WINDOWS/System32/WindowsPowerShell/v1.0:/mnt/c/Program': not a valid identifier
-bash: export: `Files/PuTTY:/mnt/c/Program': not a valid identifier
-bash: export: `Files/Git/cmd:/mnt/c/Users/dunja/AppData/Local/Microsoft/WindowsApps:/mnt/c/WINDOWS/sysnative:/snap/bin': not a valid identifier
(base) dunja@LAPTOP-5GHLIDBM:~$
UNREGISTERED VERSION - Please support MobaXterm by subscribing to the professional edition here: https://mobaxterm.mobatek.net

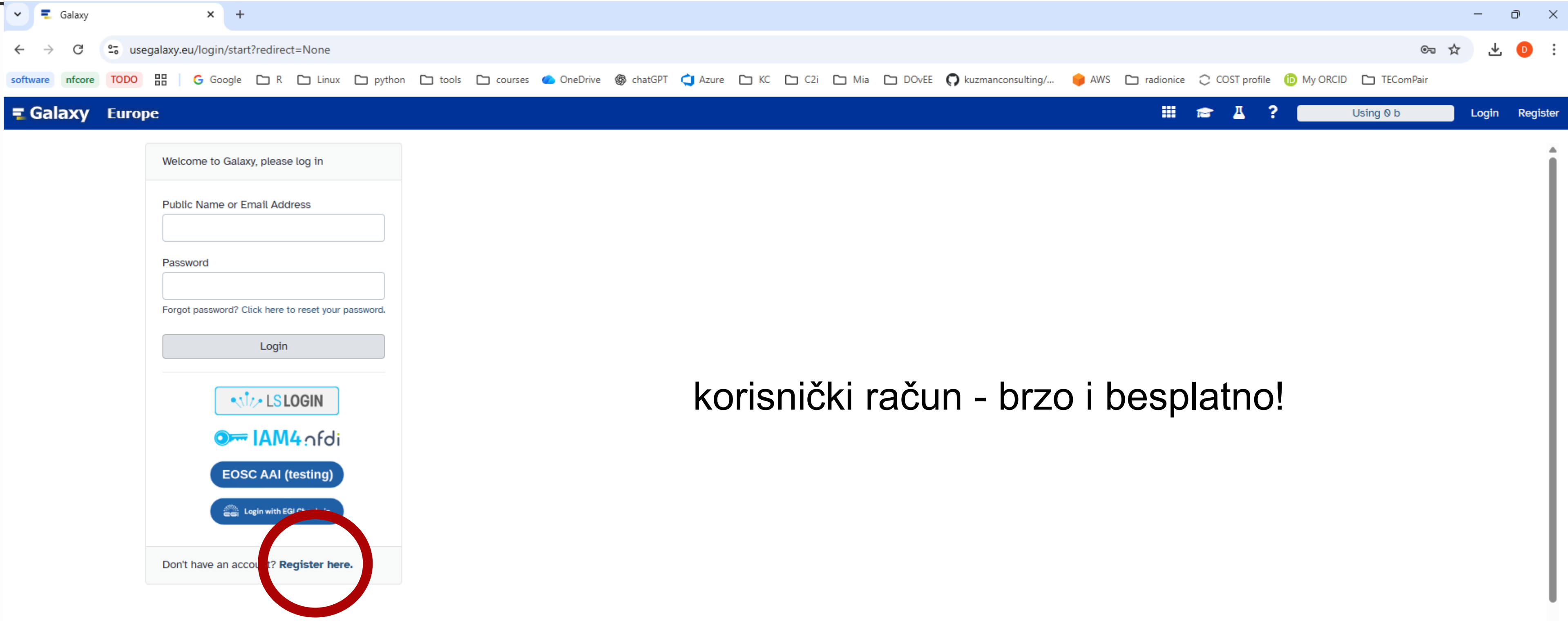

File Edit Selection View Go Run Terminal Help
projects [WSL: Ubuntu]
EXPLORER
PROJECTS [WSL: UBUNTU]
PairCompTE
nf-paircompte
modules
nf-core
repeatmodeler
subworkflows
local
nf-core
tests
workflows
.gitattributes
.gitignore
.gitpod.yml
.keep
.nextflow.log
.nextflow.log.1
.nf-core.yml
.pre-commit-config.yaml
.prettierignore
.prettierrc.yml
CHANGELOG.md
CITATIONS.md
CODE_OF_CONDUCT.md
LICENSE
main.nf
modules.json
nextflow_schema.json
nextflow.config
nf-test.config
README.md
ro-crate-metadata.json
TODO.txt
tower.yml
README.md
TEComPair
test_out
work
.nextflow.log
.nextflow.log.1
OUTLINE
TIMELINE
main.nf .../annotate_repeats
TODO.txt M
main.nf .../nf-paircompte 1
main.nf .../utils_nfcore_paircompte_pipeline
nextflow.config M
modules.config M
main.nf 3
PairCompTE > nf-paircompte > main.nf
params.outdir,
params.input
)
for now just read in samplesheet - later will include that in initialisation
_samplesheet = Channel.fromPath(params.samplesheet)
.splitCsv(header: false)
.map { row ->
def meta = row[0..-2] // all columns except last will be metadat
def fasta = file(row[-1]) // last column is expected to be file loc
tuple(tuple(*meta), fasta)
}
.map{ meta, file -> // whatever is in first column will be tal
def metaMap = [id: meta[0], rest: meta.size() > 1 ? meta[1..-1] : []]
tuple(metaMap, file) // metadata is now a Groovy map with first
}
WORKFLOW: Run main workflow
CORE_PAIRCOMPTE (
//// PIPELINE_INITIALISATION.out.samplesheet
ch_samplesheet
SUBWORKFLOW: Run completion tasks
TODO
PIPELINE_COMPLETION (
params.email,
params.email_on_fail,
PairCompTE > nf-paircompte > modules > nf-core > gffread > main.nf > process GFFREAD
process GFFREAD {
when:
task.ext.when == null || task.ext.when
script:
def args = task.ext.args ?: ''
def prefix = task.ext.prefix ?: "${meta.id}"
def extension = args.contains("-T") ? 'gtf' : ((['-w', '-x'
def fasta_arg = fasta ? "-g $fasta" : ''
def output_name = "${prefix}.${extension}"
def output = extension == "fasta" ? "$output_name" : "-o $o
def args_sorted = args.replaceAll(/(.*)(-[wxy])(.*)/) { all, pre, par
// args_sorted = Move '-w', '-x', and '-y' to the end of the args st
if ("$output_name" in ["$gff", "$fasta"]) error "Input and output
"""
gffread \\
$gff \\
$fasta_arg \\
$args_sorted \\
$output
cat <<-END_VERSIONS > versions.yml
"${task.process}":
gffread: \$(gffread --version 2>&1)
END_VERSIONS
"""
stub:
def args = task.ext.args ?: ''
def prefix = task.ext.prefix ?: "${meta.id}"
def extension = args.contains("-T") ? 'gtf' : ((['-w', '-x'
def output_name = "${prefix}.${extension}"
if ("$output_name" in ["$gff", "$fasta"]) error "Input and output
PROBLEMS 4 OUTPUT DEBUG CONSOLE TERMINAL PORTS 1
bash - projects
bash: export: `Files/WindowsApps/MicrosoftCorporationII.WindowsSubsystemForLinux_2.1.5.0_x64__8wekyb3d8bbwe:/mnt/c/Users/dunja/AppData/Roaming/MobaXterm/slash/bin:/mnt/c/WINDOWS:/mnt/c/WINDOWS/system32:/mnt/c/WINDOWS/system32:/mnt/c/WINDOWS/System32/Wbem:/mnt/c/WINDOWS/System32/WindowsPowerShell/v1.0:/mnt/c/Program': not a valid identifier
bash: export: `Files/PuTTY:/mnt/c/Program': not a valid identifier
bash: export: `Files/Git/cmd:/mnt/c/Users/dunja/AppData/Local/Microsoft/WindowsApps:/mnt/c/WINDOWS/sysnative:/snap/bin': not a valid identifier
(base) dunja@LAPTOP-5GHLIDBM:~/projects$ Open chat ctrl + I. Start typing to dismiss.
WSL: Ubuntu PairCompTE dev* ⊗ 1 ⚠ 3 1
dunja-kuzmanconsulting (2 weeks ago) Ln 98, Col 6 Spaces: 4 UTF-8 LF {} Nextflow Finish Setup Prettier

Galaxy

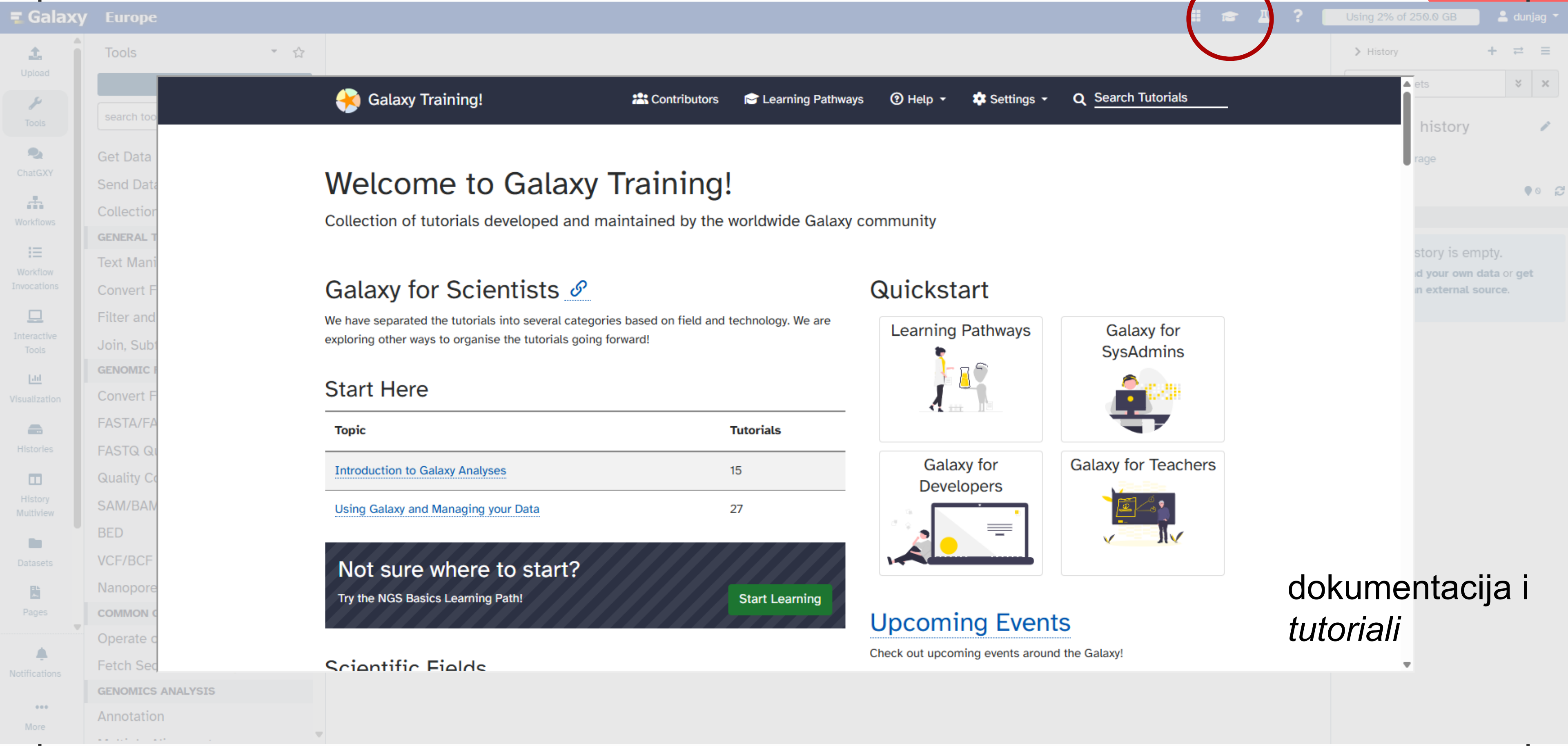
Galaxy EUROPE

- uređeni i održavani skup alata
- “ugrađene” referentne sekvence
- besplatni resursi – uz rizik čekanja

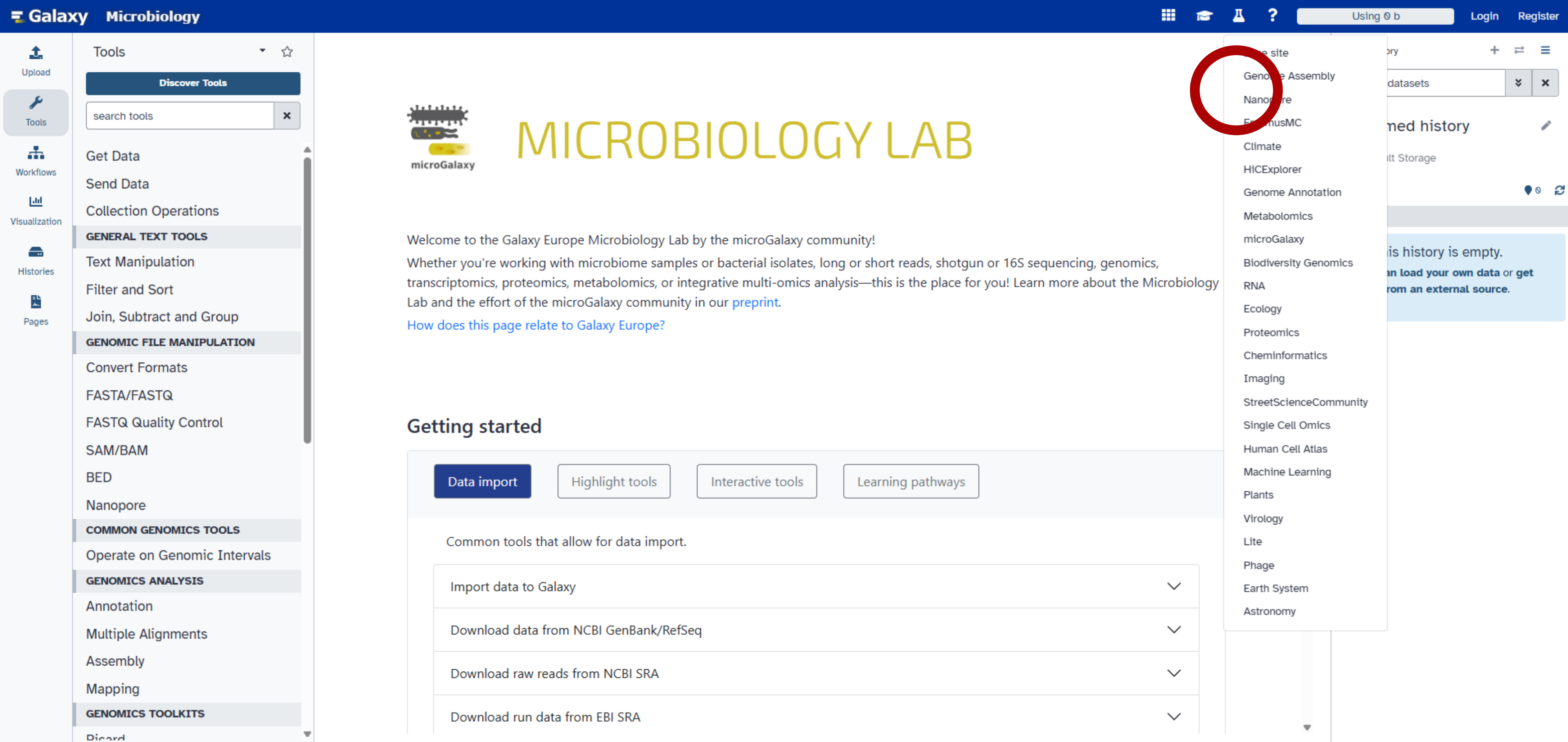




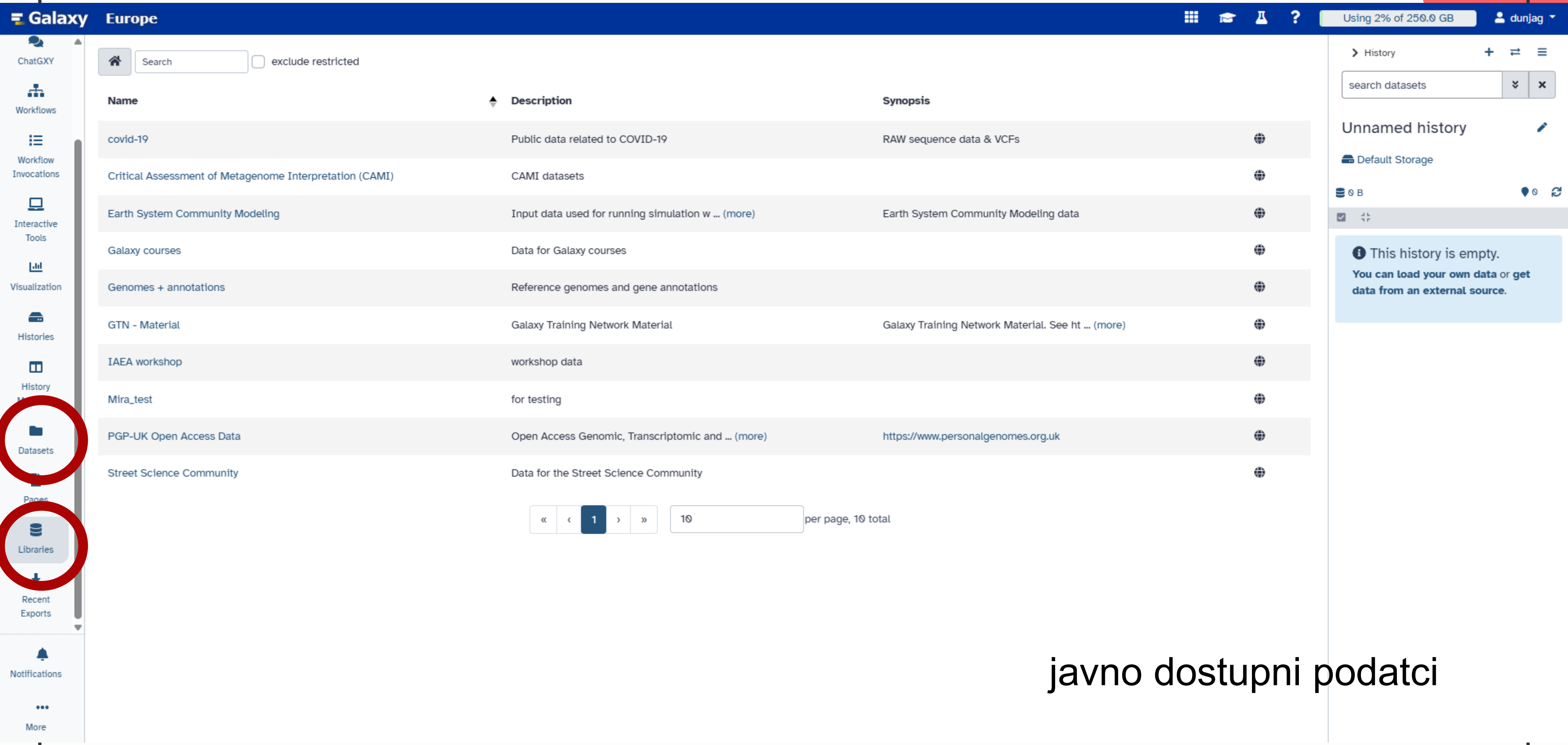
korisnički račun - brzo i besplatno!



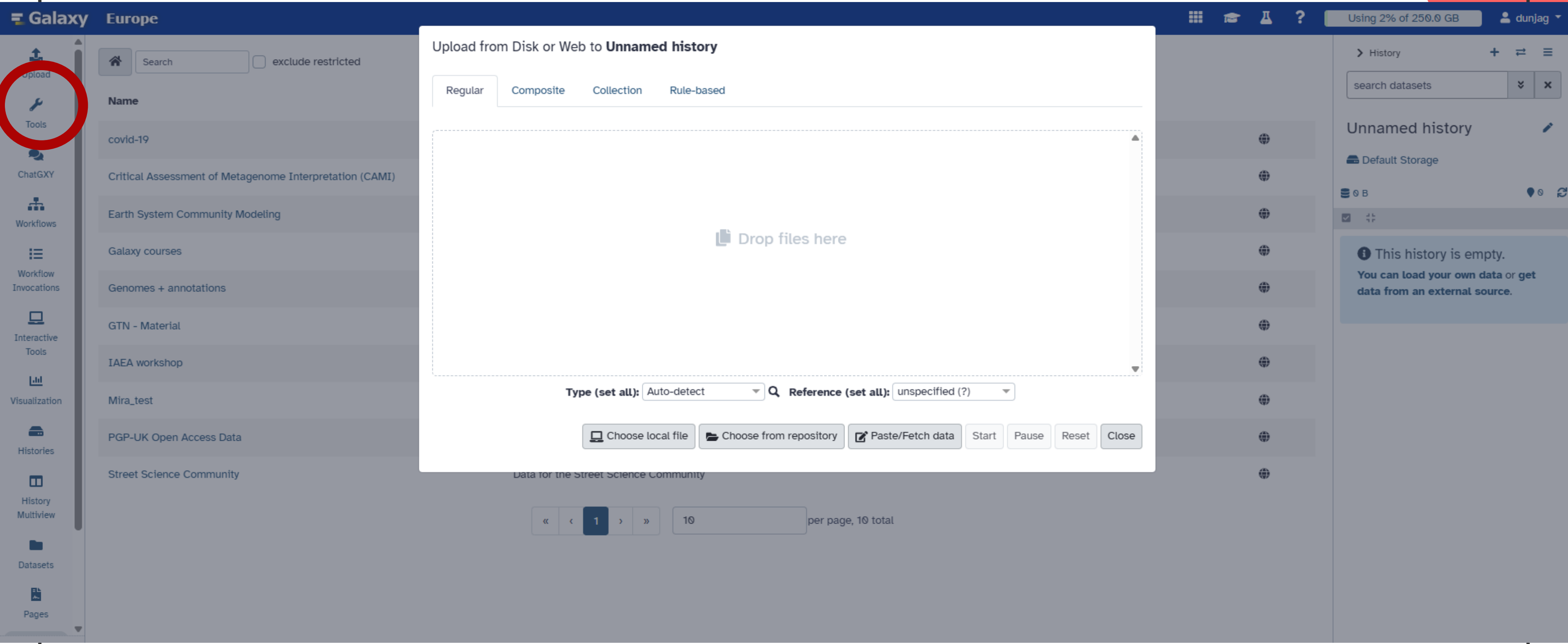
dokumentacija i *tutoriali*



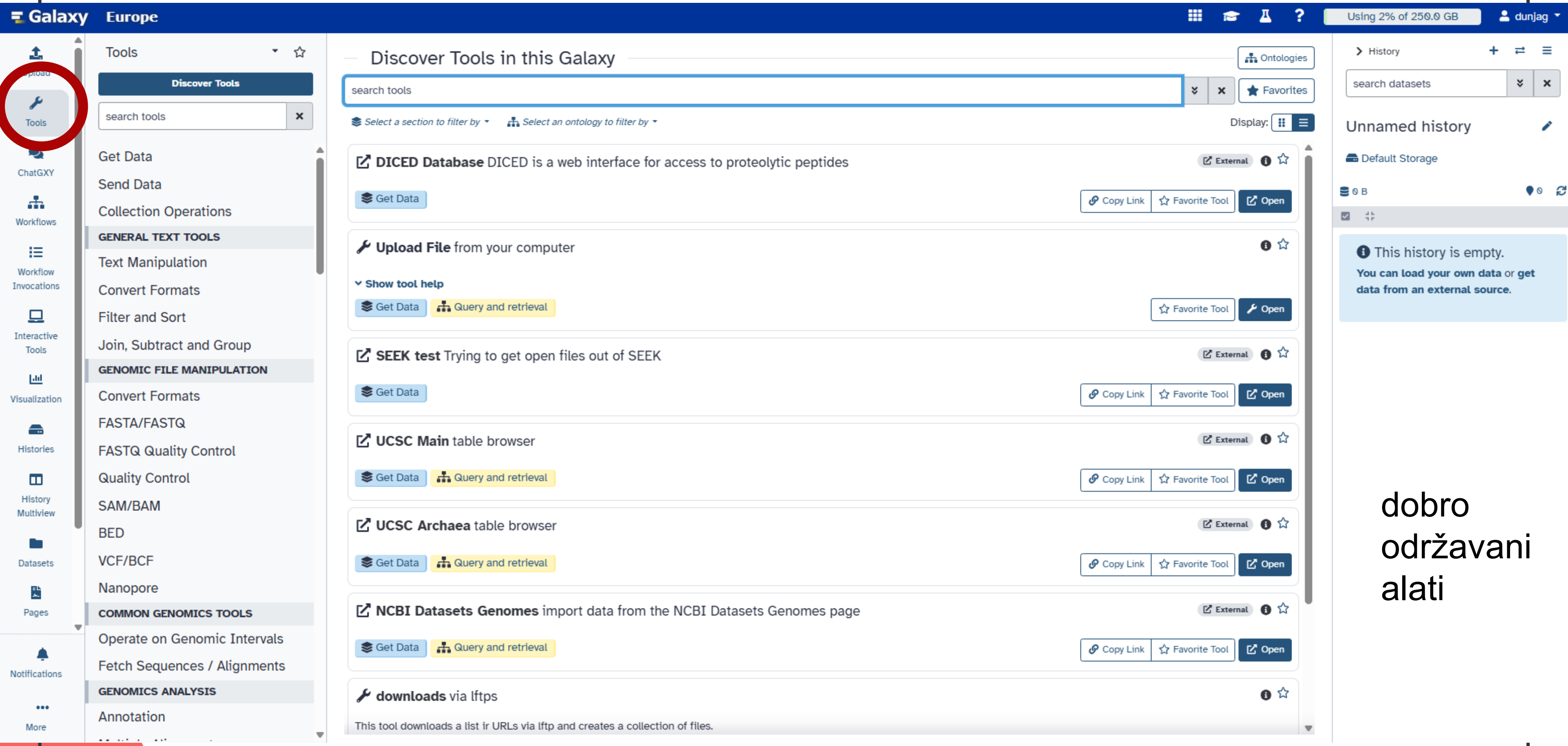
specijalizirane pod-stranice olakšavaju izbor alata



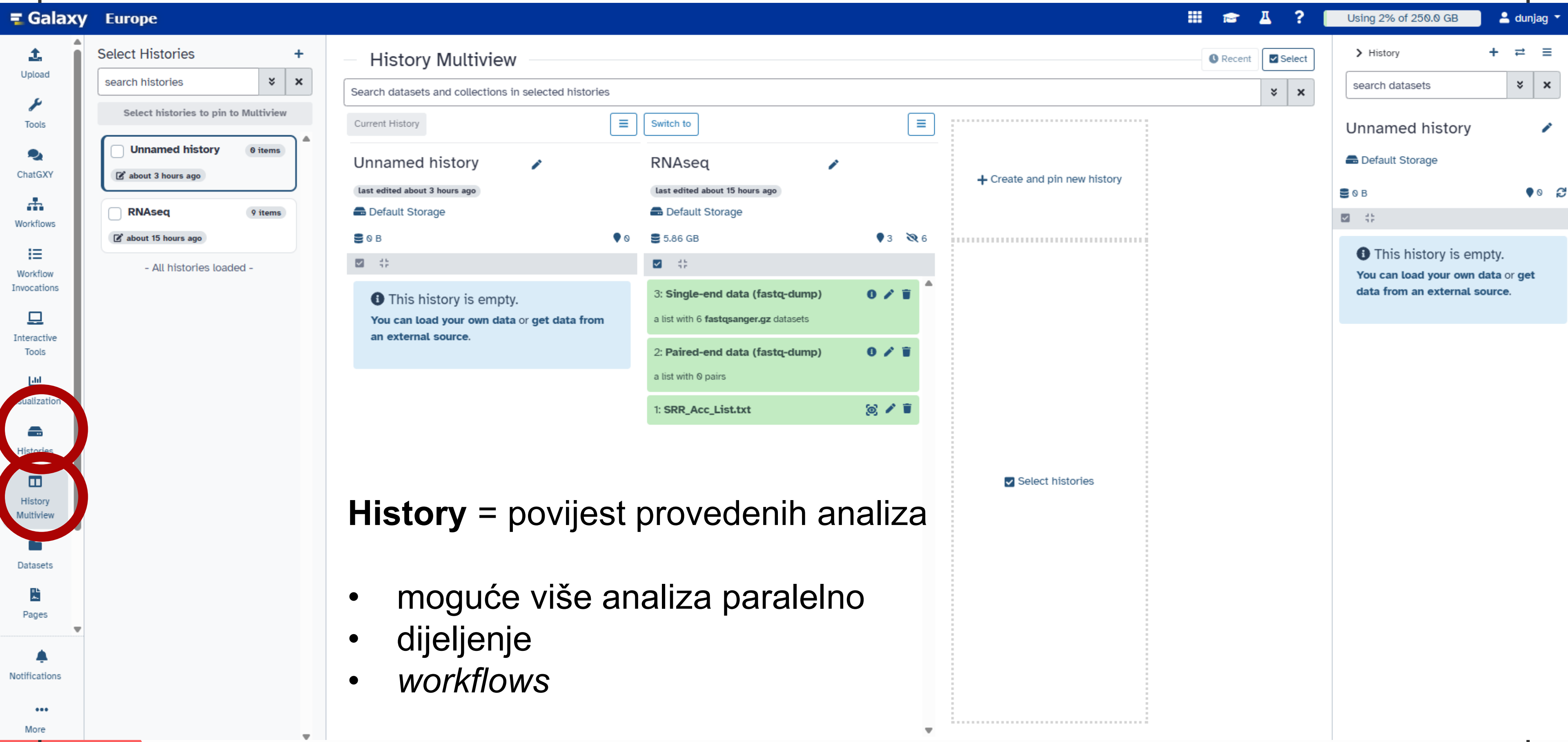
javno dostupni podatci



vaši podatci



dobro
održavani
alati



History = povijest provedenih analiza

- mogući više analiza paralelno
- dijeljenje
- *workflows*